

```
1 # conditional statements
2 # if
3 x = 20;
4 if x > 10:
5     print(f"{x} is greater than 10")
6
7 print("this outside!")
```

20 is greater than 10
this outside!

```
1 # elif, else
2 weather = "rainy"
3
4 if weather == "sunny":
5     print("Lets go out for a lunch!")
6 elif weather == "rainy":
7     print("Take a umbrella.")
8 else:
9     print("Stay indoor")
```

Take a umbrella.

```
1 # nested if statements
2 age = 20
3 has_id = True
4
5 if age >= 18:
6     if has_id:
7         print("You can enter the building!")
8     else:
9         print("You cant enter")
10 else:
11     print("You are tool young to enter!")
```

You are tool young to enter!

```
1 # multiple condition using logical operators
2 score = 80
3 attendance = 90
4
5 if score >= 80 and attendance >= 75:
6     print("Grade A")
7 elif score >= 60 and attendance >= 60:
8     print("Grade B")
9 else:
10     print("Grade C")
11
12 # Ternary opertor ( "true output" if (condition) else "false output"
13 num = 7
14 result = "Even" if num % 2 == 0 else "Odd"
15 print(result)
16
```

Grade A
Odd

```
1 # Loops
2 # loop on string
3 my_string = "Python"
4 for char in my_string:
5     print(char)
6
7 # loop on list
8 fruits = ["apple", "orange", "grapes", "cherry"]
9 for f in fruits:
10     print(f"{f} is the fruit")
11
12 # loop on tuple
13 a = (10.9, 29.9, 29, 49,7)
14 for point in a:
15     print(f"coordinates: {point}")
16
17 # loop on set
18 numbers = {1, 2, 3, 4,5 ,7}
19 for n in numbers:
20     print(f"{n} is the set number")
21
22 # dictionary
```

```

23 student_grades = {"alina": 90, "alex": 80, "alishka": 60, "alaya": 70}
24 # get the keys (default)
25 for k in student_grades:
26     print(f"{k}: is the key")
27
28 # get the value
29 for k in student_grades.values():
30     print(f"{k}: is the value")
31
32 # iterate over both keys and values
33 for st, gr in student_grades.items():
34     print(f"{st}: {gr}")
35

```

```

P
y
t
h
o
n
apple is the fruit
orange is the fruit
grapes is the fruit
cherry is the fruit
cordinales: 10.9
cordinales: 29.9
cordinales: 29
cordinales: 49
cordinales: 7
1 is the set nunber
2 is the set number
3 is the set number
4 is the set number
5 is the set number
7 is the set number
alina: is the key
alex: is the key
alishka: is the key
alaya: is the key
90: is the value
80: is the value
60: is the value
70: is the value
alina: 90
alex: 80
alishka: 60
alaya: 70

```

```

1 # loop over range
2 for n in range(0, 11):
3     print(n)
4
5 for n in range(3, 31, 3):
6     print(n)

```

```

0
1
2
3
4
5
6
7
8
9
10
3
6
9
12
15
18
21
24
27
30

```

```

1 # nested loops
2 for i in range(1, 6):
3     for j in range(1, 5):
4         print(f"{i}, {j}")
5
6 # infinite loop
7 # while True:
8 #     print("this is a infinite loop")

```

```

1, 1
1, 2
1, 3
1, 4
2, 1
2, 2
2, 3
2, 4
3, 1
3, 2
3, 3
3, 4
4, 1
4, 2
4, 3
4, 4
5, 1
5, 2
5, 3
5, 4

```

```

1 '''
2 while condition:
3     block of code
4     inc/dec statement
5 '''
6
7 countdown = 3
8 while countdown >= 0:
9     print(f"countdown: {countdown}")
10    countdown -= 1

```

```

countdown: 3
countdown: 2
countdown: 1
countdown: 0

```

```

1 # break: terminating the loop
2 for n in range(1, 10):
3     if n == 5:
4         print(f"{n} is found")
5         break
6     print(n)
7
8

```

```

1
2
3
4
5 is found
1
2
3 is found
4
5
6
7
8
9

```

```

1 # continue: terminate the current iteration
2 for n in range(1, 10):
3     if n == 3:
4         continue
5     print(n)

```

```

1
2
4
5
6
7
8
9

```

```

1 def add():
2     a = 10
3     b = 20
4     result = a + b
5     print(f"Result: {a} + {b} = {result}")
6 add()
7
8 # parameterised function

```

```
9 def mutltiply(x, y):  
10     result = x * y  
11     print(f"Result is:{x} * {y} = {result}")  
12  
13 mutltiply(2, 10)
```

Result: 10 + 20 = 30
Result is:2 * 10 = 20